Biology- II ADV
LESSON-2 U-2

The immune system

Hi everyone

I made this file to help you in your study

Hope its helpful!

Thanks for reading

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Nonspecific Immunity

These defenses are nonspecific they do not target a specific pathogen. They protect against any pathogen.
Nonspecific immunity helps prevent disease
Nonspecific immunity works quickly, but specific immunity is more effective.

Body barriers

Physical

Skin and oil Skin

Chemical

. Saliva, tears, and nasal secretions contain the enzyme lysozyme

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What if pathogens get past the barriers?

the body's nonspecific immunity continues the defense.

Cellular Defense

1-phagocytosis by which white blood cells, especially neutrophils and macrophages, engulf and absorb foreign microorganisms. The phagocytes then release chemicals that destroy the microorganisms.

2-Blood plasma contains approximately 20 complement proteins. Complement proteins aid phagocytosis by helping phagocytes

3-Interferon is a protein secreted by virus-infected cells. Interferon binds to neighboring cells and causes these cells to produce antiviral proteins. Antiviral proteins can prevent virus cells from multiplying.

4-Inflammatory Response When pathogens damage tissue, both the invading pathogen and the cells of the body release chemicals. These chemicals attract phagocytes, increase blood flow to the infected area, and make blood vessels more, Pain, heat, and redness in the infected area are the result of the inflammatory response.

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Specific Immunity

When pathogens get past the nonspecific defenses, the body has a second line of defense the lymphatic system.

Specific immunity is more effective but takes more time to develop.

the functions of the lymphatic system

1-filters lymph and blood and destroys foreign microorganisms.

2-Lymph leaks out of capillaries to bathe body cells.

3-After the fluid circulates among the cells, lymphatic vessels collect and return it to the veins near the heart.

part of the lymphatic system

lymphatic system contain lymph, lymphocytes, a few other cell types, and connective tissue.

Lymphocytes are a type of white blood cell that is produced in red bone marrow.

The lymph nodes are located in different places along the body, filter the lymph and remove foreign materials.

The tonsils form a protective ring between the nose and mouth.

The spleen stores blood and destroys damaged red blood cells.

The thymus gland, located above the heart, helps to activate a special kind of lymphocyte called a T cell. T cells are produced in the bone marrow, but they mature in the thymus gland.

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B Cell Response

When foreign substance, enters the body, it triggers the production of antibodies. Antibodies are proteins produced by B lymphocytes that specifically react with the antigen. B cells, are antibody factories located in all lymphatic tissues.

1-When a macrophage engulfs a pathogen, it displays it as a processed antigen

2-the macrophage, with the process antigen ,binds to type of lymphocyte called a helper T cell. This process activates the helper T cell

3-activated helper T cell then activates B cells and another type of T cell to produce antibodies

4-Antibodies bind to the microorganisms, making phagocytosis more likely. They can also trigger an inflammatory response.

5-helper T cells can activate lymphocytes called cytotoxic T cells.

6-cytotoxic T cells destroy pathogens and release chemicals called cytokines. Cytokines stimulate cells of the immune system to divide and attract immune cells to the infected area.

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Memory cells

are long-living cells that are exposed to the antigen during the primary immune response.

If the body is later exposed to the same pathogen, memory cells respond rapidly to protect the body.

Passive and Active Immunity

This type of temporary protection occurs when antibodies are made by other people or animals.

EX. mother and child

These antibodies are injected into people who have been exposed to that particular infectious disease.

Antibodies also are available to inactivate snake and scorpion venoms.

Active immunity occurs after the immune system is exposed to disease antigens and memory cells are produced.

Immunization, also called vaccination, is the deliberate exposure of the body to an antigen so that a primary response and immune memory cells will develop

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Immune System Failure

Some diseases reduce the immune system's effectiveness. One such disease, acquired immunodeficiency syndrome (AIDS), results from infection by human immunodeficiency virus (HIV). HIV infects helper T cells, turning them into HIV factories. The new viruses then infect other helper T cells. Over time, loss of helper T cells reduces the body's ability to fight off diseases. Patients might exhibit few symptoms for as many as ten years but can still spread the infection through blood.

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